

BERGEN, Germany

WMO: 102380

Lat: 52.8153N

Lon: 9.9247E

Elev: 254

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 97-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	18.2	10.1	9.3	16.1	14.4	11.5	20.2	14.7	45.5	13.4	46.5	3.3	80	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	85.5	66.8	81.7	65.3	78.1	63.9	68.5	80.5	66.9	78.4	65.4	75.7	5.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	92.0	71.9	63.0	86.9	70.8	61.5	82.4	69.2	32.8	80.4	31.5	78.1	30.3	76.2	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.0	9.1	DB	6.6	92.8	7.1	3.8	1.5	95.5	-2.6	97.7	-6.6	99.9	-11.8	102.6	(4)
(5)			WB	7.7	71.7	7.2	1.9	2.6	73.1	-1.6	74.2	-5.7	75.3	-10.9	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	34.0	34.8	39.7	48.0	55.2	60.1	63.8	63.7	57.5	48.7	41.0	34.2		
	DBStd	12.89	8.09	7.44	7.02	6.97	6.17	6.07	5.87	5.46	5.53	6.49	6.53	7.93		
	HDD50	2257	495	427	322	119	20	2	0	0	4	103	274	491		
	HDD65	6180	960	845	783	509	309	171	97	93	233	505	719	956		
	CDD50	1698	0	1	4	60	181	305	426	424	229	63	5	0		
	CDD65	147	0	0	0	0	5	24	58	52	8	0	0	0		
	CDH74	1822	0	0	0	10	140	315	692	547	116	2	0	0		
	CDH80	511	0	0	0	0	23	74	226	172	16	0	0	0		
(14) Wind	WSAvg	4.2	4.8	4.7	4.7	4.2	4.0	3.9	4.0	3.7	3.5	4.0	4.1	4.5		
(15) Precipitation	PrecAvg	30.5	3.0	2.2	2.2	1.8	2.3	2.7	3.1	2.9	2.5	2.4	2.6	2.9		
	PrecMax	41.9	5.7	4.9	5.0	3.6	5.5	5.0	7.5	5.5	7.8	7.4	5.1	6.7		
	PrecMin	21.7	0.2	0.6	0.3	0.3	0.8	1.1	0.8	1.1	0.6	0.7	0.1	0.9		
	PrecStd	5.1	1.5	1.1	1.3	0.8	1.1	1.1	1.6	1.3	1.6	1.3	1.3	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	55.5	63.8	75.1	83.1	86.7	91.4	90.5	82.6	68.7	58.7	52.6		
		MCWB	50.1	49.0	51.6	58.1	63.5	67.2	67.5	68.2	65.3	61.0	53.2	49.7		
	2%	DB	50.8	51.2	59.5	70.6	77.8	81.6	86.3	84.8	77.5	64.8	55.4	49.6		
		MCWB	47.8	47.0	49.4	55.9	60.9	65.0	67.3	66.8	63.6	58.1	51.3	47.0		
	5%	DB	47.9	48.5	54.8	66.5	73.5	77.3	81.6	80.2	72.6	61.5	52.6	47.1		
		MCWB	45.4	45.2	48.1	53.4	58.9	63.2	65.6	66.1	61.5	56.0	49.5	45.0		
	10%	DB	45.4	45.7	51.0	62.0	69.4	73.3	77.2	75.8	68.4	58.7	50.3	44.6		
		MCWB	43.4	42.9	46.1	52.0	56.9	61.2	64.0	63.9	59.8	54.5	47.6	42.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	50.2	53.9	59.9	66.5	68.9	71.0	71.0	67.2	61.9	54.9	50.3		
		MCDB	52.4	53.1	60.7	72.0	79.7	81.4	84.9	84.7	77.6	68.0	57.4	51.6		
	2%	WB	48.1	47.9	51.1	57.0	62.8	67.0	68.4	68.5	64.6	58.9	52.0	47.5		
		MCDB	50.6	50.5	56.2	67.1	74.3	78.2	80.9	79.8	74.5	63.3	54.6	49.3		
	5%	WB	45.6	45.5	48.8	54.8	60.2	64.8	66.7	66.7	62.6	56.8	49.8	45.0		
		MCDB	47.4	48.0	53.7	63.7	70.4	74.8	79.0	77.2	70.6	60.7	52.2	46.7		
	10%	WB	43.5	43.0	46.4	52.7	57.9	62.5	65.1	65.0	60.8	54.8	47.9	42.9		
		MCDB	45.3	45.6	51.1	60.7	67.0	70.3	75.4	74.1	67.1	58.1	49.9	44.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	10.4	14.4	19.5	20.5	20.1	18.8	19.0	18.9	14.9	9.6	8.5		
		MCDBR	10.2	13.0	22.5	27.9	29.3	27.3	27.8	27.0	26.3	17.8	11.9	9.6		
		MCWBR	8.5	9.4	14.0	14.7	14.6	12.9	11.6	11.8	13.9	11.7	8.9	8.5		
	5% WB	MCDBR	9.3	10.6	18.6	23.3	25.3	24.2	24.7	22.5	22.5	16.4	11.0	9.3		
		MCWBR	8.4	8.3	12.6	13.0	13.8	12.4	11.3	10.9	12.4	11.3	9.0	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.336	0.369	0.410	0.409	0.407	0.424	0.417	0.385	0.354	0.334	0.310		
	taud		2.369	2.362	2.292	2.218	2.262	2.287	2.251	2.286	2.363	2.435	2.409	2.360		
	Ebn at Noon		212	244	257	260	267	268	261	255	250	234	203	191		
	Edh at Noon		19	25	33	40	40	40	40	37	30	23	18	16		
(44) All-Sky Solar Radiation	RadAvg		196	415	764	1260	1557	1658	1574	1319	964	545	241	151		
	RadStd		20	58	80	153	175	129	214	102	114	56	26	12		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-256	N/A	N/A

Nomenclature: See separate page